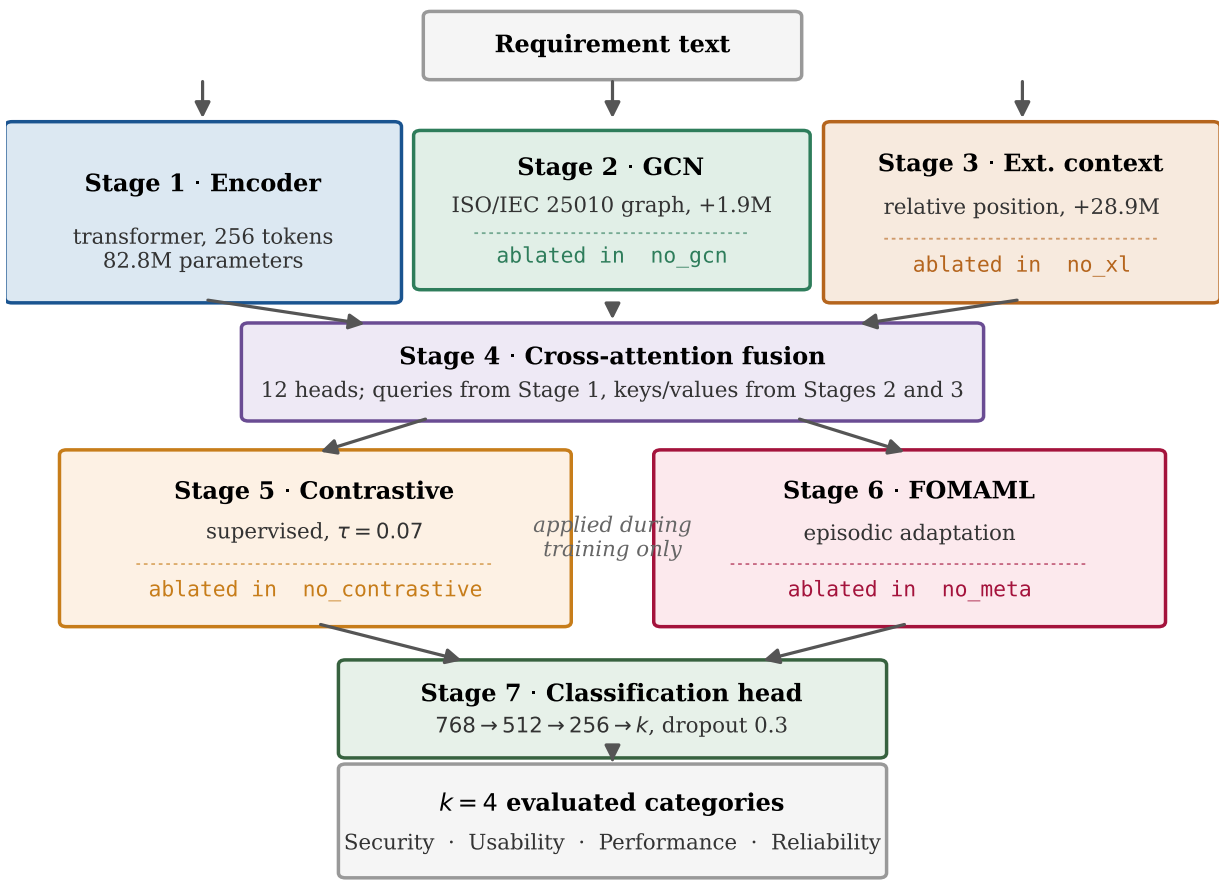
Requirement text
Stage 1 · Encoder
transformer, 256 tokens
82.8M parameters
Stage 2 · GCN
ISO/IEC 25010 graph, +1.9M
ablated in no_gcn
Stage 3 · Ext. context
relative position, +28.9M
ablated in no_xl
Stage 4 · Cross-attention fusion
12 heads; queries from Stage 1, keys/values from Stages 2 and 3
Stage 5 · Contrastive
supervised, $\tau = 0.07$
ablated in no_contrastive
applied during training only
Stage 6 · FOMAML
episodic adaptation
ablated in no_meta
Stage 7 · Classification head
$768 \to 512 \to 256 \to k$, dropout 0.3
$k = 4$ evaluated categories
Security · Usability · Performance · Reliability